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A Rate-Dependent Upscaling Framework for REV Saturation Functions in Dual-Porosity Media

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Summary

Simulating multi-phase flow at length scale larger than measured scale has been an active topic for decades. In this study, we address the numerical challenges associated with estimating effective relative permeability and capillary pressure (usually known as saturation functions) of dual porosity systems. Specifically, we introduce a modified version of periodic boundary conditions applied in steady-state upscaling workflow. Our modification is based on mirror-periodic boundary conditions as a reasonable choice overcoming short failings introduced through direct application of periodic boundary conditions. This modification can be seamlessly coupled with previous approaches handling force balance spectrum from capillary limit to viscous limit. In addition, the resultant relative permeability is a tensor allowing for modeling complex geometrical configurations. The approach is demonstrated through estimation of effective multi-phase parameters of bioturbated strata model.

Introduction

Predicting recovery factor (RF) or ultimate storage capacity (USC) is critical for assessing the success of subsurface processes such as hydrocarbon production and geological gas storage. Theoretically, these metrics are governed by

$$RF \quad (or \quad USC) \quad = SE * DE \tag{1}$$

where *SE* is sweep efficiency, and *DE* is displacement efficiency (Lake et al., 2014). Eq. (1 indicates that both *SE* and *DE* contribute equally to RF/USC. From a numerical point of view, despite continuous advances in computational capabilities, explicitly modelling of natural nested heterogeneities—spanning micrometer to kilometer scales—remains infeasible (Wen et al., 2003). Instead, these heterogeneities are typically lumped during grid coarsening (King, 2007). This process artificially inflates *SE* increases as a result of grid dimensions increase. Hence, *DE* shall compensate for *SE* inflation and sub-grid heterogeneity. Estimating petrophysical parameters at a larger scale than the input scale is referred to as upscaling. If done correctly, upscaling shall preserve the effect of small-scale heterogeneities.

Ringrose et al. (1993) classified Darcy-scale heterogeneity into three categories by length scale: lamina (mm-cm), bedform (cm-m), and formation (m-10s of m). Here, we focus on upscaling from the lamina to the bedform scale. Accurate upscaling hinges on properly representing saturation functions (relative permeability and capillary pressure), which dictate *DE*. While recent laboratory experiments have explored multi-phase flow in meter-scale cores (e.g., Seyyedi et al., 2022), such studies remain limited in scope and applicability.

Excluding experimental analysis, the literature shows that existing multi-phase upscaling approaches can be grouped into two categories: 1) analytical approaches, and 2) numerical approaches. The former is simple and efficient (Rabinovich et al., 2016). However, its applicability is limited to force balance endmembers (discussed later in detail) and requires empirical tuning parameters for complex geological settings (cf., Ekrann et al., 1996). On the other hand, numerical approaches are flexible in handling different geological representations (Boon et al., 2022), however, they are sensitive to selected boundary conditions as demonstrated later. In this study we consider numerical upscaling with a reasonable treatment of boundary conditions eliminating boundary artifacts.

Flow based numerical upscaling goes through two sequential steps: 1) saturation distribution determination (via steady-state or unsteady-state simulation), and 2) estimating ensembled parameters (Virnovsky et al., 2004). Unsteady-state methods better reflect actual displacement but can yield non-physical relative permeability values (e.g., negative or >1; Barker and Dupouy, 1996). On the other hand, steady-state methods produce well-behaved curves but are highly boundary-condition-dependent (Ekrann and Aasen, 2000).

Following this brief introduction, this article is structured into six sections: Modeling background, Methodology, Bioturbated model, Results, Discussion and Conclusion. Modeling background spots the light on theoretical background of force balance, and different boundary conditions applied in steady-state upscaling. Methodology section details the proposed framework of upscaling and the application of mirror-periodic boundary conditions. Then, Bioturbated model describes geological and petrophysical parameters of bioturbated REV model utilized as demonstration example of our framework. Results section presents upscaled saturation functions of bioturbated model. After that, discussion section clarifies why mirror-periodic boundary conditions is capable of bridging the gap between engineering and geological description of REV specially for two-phase flow. Finally, the conclusion section summaries key findings out of this study.

Modeling background

Capillary interfaces and rate-dependency

Consider two-phase flow (wetting and non-wetting phases) through a heterogeneous medium comprising two rock types: a coarse rock characterized by relatively lower capillary pressure and a fine rock has higher entry pressure, as illustrated in Fig. 1a. The flow takes place from coarse rock toward fine rock. Van Duijn et al. (1995) showed that the interface between these rocks acts as a capillary barrier, trapping the non-wetting phase until the fine rock's entry pressure is exceeded. This creates a saturation discontinuity—while capillary pressure equilibrates across the interface, the saturation profile exhibits a jump. On the upstream side, capillary forces diffuse the saturation profile away from interface until equilibrating with viscous forces (advection) is achieved. The extension of this diffusive layer is rate-dependent: higher pressure gradients (stronger viscous forces) compress the layer, while lower gradients allow it to expand as shown in Fig. 1b. This layer plays a significant role controlling effective parameters, especially when upscaling from lamina scale to bedform scale where size of such layer is comparable to correlation length of heterogeneity (Lohne et al., 2006).

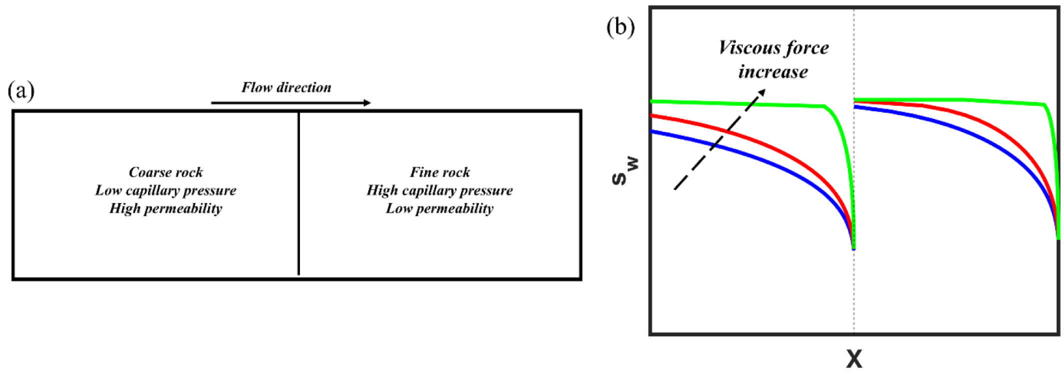

Figure 1—(a) Schematic model used to explain the role of capillary jumps. The model shows two phases moving from coarser material in the direction of finer material. (b) Influence of force balance on capillary spreading at upstream side of interface. The increase in viscous force (from blue to green) compresses diffusive layer and minimizes its influence on effective parameters. The interface is shown by dashed black line.

Boundary conditions

To figure out steady-state saturation, two different boundary conditions can be applied. The first option is constant fractional flow which is analogous to experimental relative permeability measurements. In this case, two-phases are injected simultaneously from one end while the other end is kept at constant pressure (e.g., Boon et al., 2022; Boon and Benson, 2021). However, Ekrann and Aasen (2000) proved that this selection of boundary conditions produces a transition zone adjacent to inlet boundary. The prediction of the transition zone size is difficult, and we may end with transition zones extending for several meters covering whole computational domain. In such scenario, the estimated effective parameters are not accurate.

The second option is periodic boundary conditions (e.g., Dale et al., 1997; Virnovsky et al., 2004). In this case, the model is initialized with certain average saturation. Then, fluids are circulated across opposite boundaries by applying certain pressure gradients (Youssef et al., 2024). The circulation continues till steady-state is achieved. A clear artifact is the artificial capillary barriers introduced via wrapping opposite boundaries when dissimilar rock types meet. In this study, we introduce the mirror-periodic boundary conditions as an alternative approach of periodic boundary conditions for the sake of maintaining benefits of periodic boundary conditions while avoiding artificial capillary barriers.

Methodology

Consider a heterogeneous domain Ω composed of at least two different facies distinct by their capillary pressure similar the one shown in Fig. 1a. To estimate steady-state saturation distribution, we start by determining flow direction, in this case x-direction as an example. Then, a reflected image of the model along y-axis is assembled with original model to form extended model Ω^e as shown in Fig. 1b. The steady-state saturation distribution is determined by solving immiscible-incompressible two-phase flow mass conservation equations

$$\varnothing \frac{\partial s_a}{\partial t} = \nabla . \left(\frac{k_{ra} k}{\mu_a} \nabla p_a \right) \qquad in \left(t > 0 \right) \cap \Omega^e \tag{2}$$

where $\varnothing$ is porosity, s is saturation, α refers to phase, t is time, k_r is relative permeability, k is absolute permeability, μ is viscosity and p is pressure. The domain is initialized with any saturation distribution that gives certain average saturation. Usually, the initial distribution is estimated from capillary limit. The boundaries along flow direction are subjected to Neuman no flow boundaries while boundaries perpendicular to flow directions are wrapped around to assure periodicity.

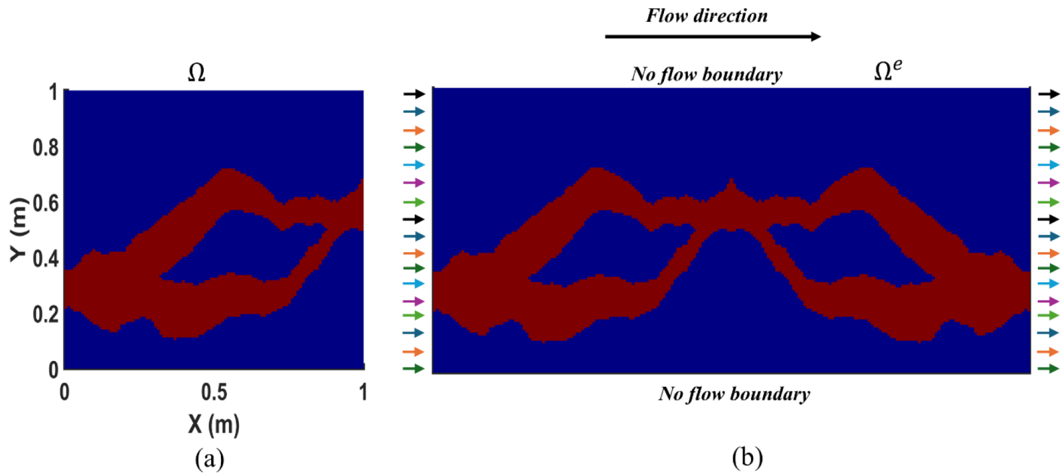

Figure 2—(a) 2D bioturbated model used as an illustrative example of proposed upscaling workflow (more details in following section). The model is composed of two rock types: high permeability burrows shown in red surrounded by low permeability host rock matrix shown in blue. (b) application of mirror periodic boundary conditions for flow along x-direction. Arrows show periodic flux along left and right boundaries. The colors indicate that this flux is variable along periodic boundaries and not necessarily constant.

Bioturbated model

We demonstrate our upscaling framework using bioturbated strata, an example of dual-porosity systems. Bioturbation—a biological process driven by organism activity—creates heterogeneous rocks by altering the petrophysical properties of the host matrix. The resulting altered regions, called burrows, can either enhance or degrade overall rock quality, with permeability contrasts between burrows and the matrix spanning a wide range (10^{-3} to 10^3).

3D model

For this study, we analyze a 3D model (Fig. 3) generated via multipoint statistics (Eltom, 2024). The model is a 1 m^3 cube where burrows constitute 39.5% of the volume, with over 90% connected as a single network. The burrows exhibit a complex dendritic morphology, preferentially oriented along z-axis as shown in Fig. 3b. To simplify analysis, we use a binarized version of the model, assigning distinct petrophysical properties to burrows and the host matrix. Table 1 summarizes these parameters for both facies.

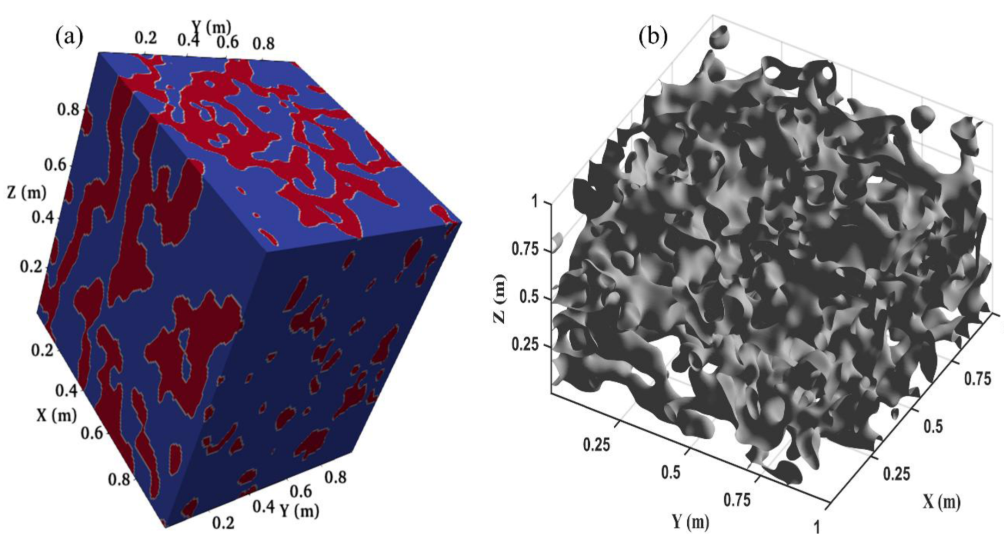

Figure 3—Three-dimensional REV of the bioturbated model used in this study. (a) Facies distribution: burrows in red and host rock matrix in blue. (b) Interface separating burrows from the matrix,

Table 1—Single- and two-phase parameters of burrows and host rock matrix.

Facies	Permeability (mD)	Porosity	Entry pressure (Pa)	Residual water saturation	Brooks-Corey parameter	Endpoint relative permeability of non-wetting phase
Host rock matrix	100	0.28	1000	0.15	2	0.95
Burrows	1000		316.228			

2D synthetic model

Without the loss of generality, instead of analyzing 3D model directly, we start by analyzing a 2D synthetic section (Fig. 2a). The aim of this analysis is to simplify the proposed approach. The 2D section is composed of a single burrow with two branches surrounding an island of low permeability host rock matrix. The burrow forms a continuous path extending from left to right boundary, and surrounded by lower permeability host rock matrix. This model is selected intentionally as boundaries of high permeability burrow do not overlap on opposite sides. The petrophysical parameters are the same as 3D model.

Results

We start by analyzing 2D synthetic model, then move to 3D model. For each model, the single-phase upscaling is presented first, then the multi-phase upscaling follows. The simulation is performed using Steady-State Upscaling module (SINTEF, 2013) embedded into MATLAB Reservoir Simulation Toolbox (MRST) (Lie and Møyner, 2021).

2D synthetic model

Fig. 4 shows the upscaled permeability tensor of the 2D model. As intuitively expected from Fig. 2a,the tensor does not align with system axes, rather it is rotated by 8° anticlockwise. The maximum eigen value of the tensor is 273.8 mD while the minimum one is 148.1 mD. The diagonal components of relative permeability at CL are shown in Fig. 5. The figure shows the critical saturation of non-wetting does not coincide in both directions. In x-direction, the critical saturation remains zero similar to one of input k_r while in y-direction it jumps to ~0.2. This variation is justifiable as burrows forms continuous pathway from left to right boundaries while there no such connectivity between top and bottom boundaries. In case if the periodic boundaries are applied solely, the connectivity existing in x-direction will be lost and critical saturation increases to the same value anticipated in y-direction.

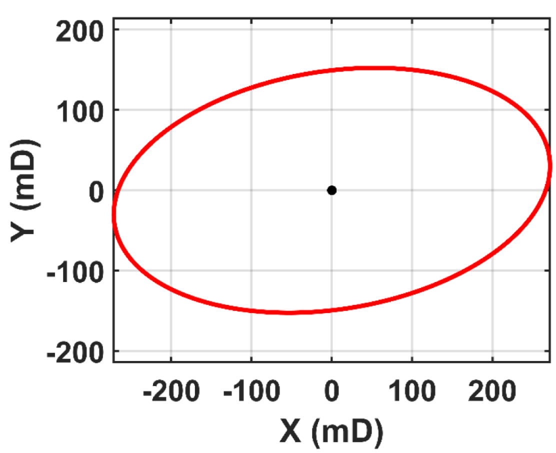

Figure 4—Absolute permeability tensor of 2D synthetic bioturbated model. The upscaled permeability is represented as an ellipse centered around origin. The maximum principal axis is 273.8 mD and titles on x-axis by 8° measured in ani-clockwise direction. The minimum principal axis is 148.1 mD. The principal axes are perpendicular to each other.

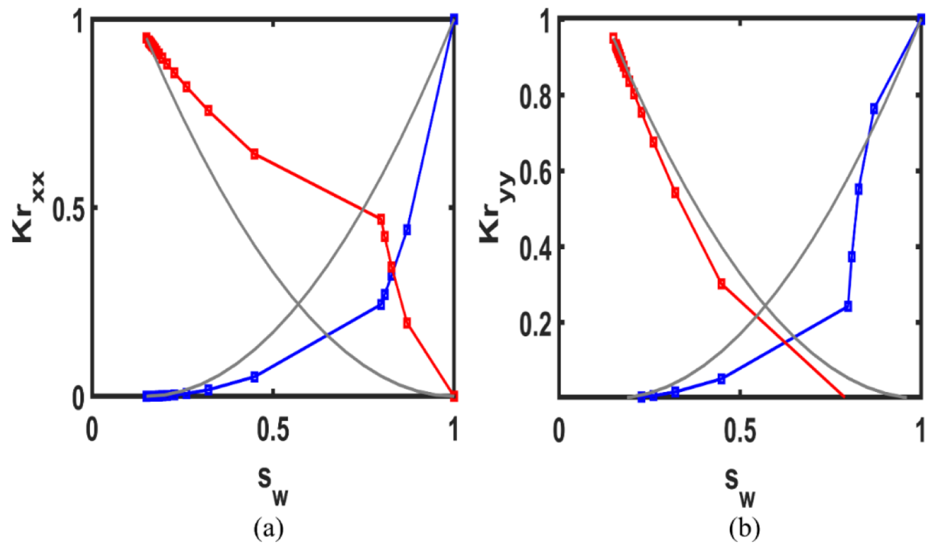

Figure 5—Upscaled relative permeability diagonal component of 2D model: (a) xx-component, and (b) yy-component. The wetting phase is shown by blue and non-wetting by red. The input relative permeability of both facies is shown by solid gray lines. The input relative permeability coincides with VL relative permeability.

Discussion

Real geological models are usually characterized by complex heterogeneities spanning different orders of length scales. Transferring these models to their counterparts digital twins includes some simplifications leading to deviation from real behavior to some extent (Mikes et al., 2006). One of the sources behind these simplifications is the restrictions exist in current modeling approach. One of these restrictions is the selection of reasonable boundary conditions. From a classical point of view, REV size occurs when a certain parameter of interest starts to stabilize dominating perturbations introduced by sub heterogenies (Bear, 1989). In quantitative manner, the factors influencing parameters of interest become periodic (e.g., Mankowski functions, Armstrong et al., 2019; Sadeghnejad et al., 2023). However, the periodicity of these factors does not necessary mean exact repetition of geometrical structure of domain of interest (cf. Trevisan et al., 2017). Hence, the application of periodic boundary conditions, although it seems most relevant, requires some idealization of facies geometry (cf.,Youssef et al., 2024). Our analysis is built on the concept of fixing geometrical representation of geological REV. Therefore, the geo-statical model built of REV is utilized directly without any modification.

The upscaling approach presented here is an extension of the recently introduced method by Youssef et al. (2024). They revealed that the periodic boundary conditions are relevant to be coupled with oriented pressure gradient field. This approach has power of modeling real pressure fields even if that are not aligned with system axes. However, their analysis was restricted to perfectly periodic domains where similar facies on opposite boundaries match each other. It is clear that violating this assumption leads to introducing capillary discontinuity accumulating non-wetting phase in lower capillary pressure medium. In our approach, boundary mirroring avoids termination of high permeability features across periodic boundaries. Hence, it overcomes boundary artifacts that is anticipated by direct application of periodic boundary conditions.

Although the usage of k_r Upscaling Module in MRST provided a fast way of applying the concept mirror-periodic boundary condition with minimum effort, the user shall be aware that this module is only valid for capillary pressure modeled using J-function. Such limitation does not put a constraint on our methodology, rather on the module itself.

Conclusion

In this study, we developed a numerical algorithm for estimating relative permeability and capillary pressure while honoring geological representation.

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